Sustainable ,Biodegradable, and synthetic, Biologically derived Immersive Bio-Coolant

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ALL OVER THE HEADLINES

Temporarily banning data centers draws more interest from state, local officials

Moratoriums would give state legislators more time to determine whether to continue investing in the massive facilities.

REUTERS

India's HCLTech to invest $1.48 billion in AI dat center, open 5,000-seat tech hub

Google to Build $15B AI Data Center Hub in India

The new data center hub will anchor Google's long-term expansion in India, linking local R&D and global co meet rising demand for cloud and AI capacity.

AI Data Center Boom Strains Global Construction Capacity

by Nathan Eddy

TECH NEWS

Data center opponents have blocked or delayed projects worth nearly $130 billion in 2026, study finds

India's data center capacity to surge 66 percent by 2026 – JLL

Industry is set to add 604MW of capacity and see an investment of $3.8 billion

November 29, 2024 By: Niva Yadav Have your say

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India's Data Centre Infrastructure Reaches 1.8 GW with Rising Global Investments; Savills India Report

Savills India Report highlights India's data centre growth with 258 MW IT capacity addition in H1 2026 driven by hyperscalers, AI, cloud adoption and digitalisation

Construction & Site Development · 2 Min Read

STT Global Data Centres acquires Navi Mumbai land for ₹282 crore expansion

India's Adani Group To Invest $100 Billion In AI Data Centers Amid Strategic Partnership With Google, Microsoft

New York becomes first U.S. state to impose AI data center ban

Community Backlash to AI Data Centers Is Growing Across the US

HOME > NEWS > THE DATA CENTER CONSTRUCTION CHANNEL

Tata acquires land in India for OpenAI data centers – report

Water Footprint of Data Centers

A medium-sized data center can consume **100 million gallons** of water annually for cooling, equivalent to the yearly water use of **1,000 households**.

About use water-based cooling as their primary cooling method.

Around 57% of data centers' water comes from **potable supplies**, with 80–90% sourced from freshwater (rivers and lakes).

But isn't just limited to water, resource it exhausts is POWER

In the U.S., data centres already consume **4–4.4%** of total electricity (2023)

Data centre power capacity in India is experiencing exponential growth, nearly **tripling from 520 MW in 2020 to 1.5 GW** by mid 2025.

Overall share of data centres to global power usage projected to rise to **2.5–3% of global electricity by 2030**



Data Centres

Ireland’s data centre nightmare – and what others can learn from it

Critics say Ireland's data centre industry has exorbitant power needs, undermines its emissions targets – and that Europe should beware.

Heidi Vella February 18, 2025

Ireland’s data centers use ~1/5 of the country’s electricity, heading to 1/3 by 2026.

- That’s almost as much power as every home in Ireland combined.
- Grid got so strained that new data centers were **banned** near Dublin in 2021.
- Demand kept rising anyway, govt spent ~€1 billion on emergency generators.
- Regular households now pay almost double what data centers pay for power.

Why do data centres require so much water and power? Blame the method

A majority of large, modern data centers use **water-based cooling** for better energy efficiency. This often involves cooling towers or evaporative chillers: warm water absorbs heat from servers and is then cooled by **evaporation** in a tower. As water evaporates into the air, it carries away heat

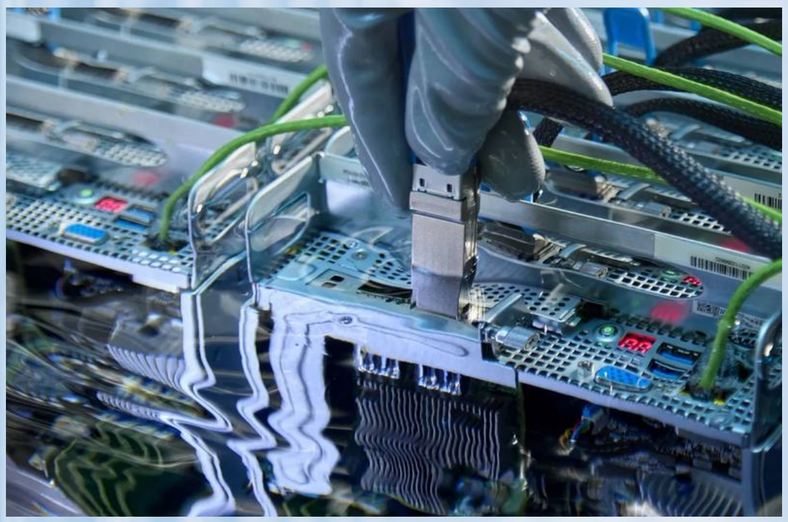


75–90%

Most data centers do use water in some form, but not all. Industry estimates show that roughly **75–90%** of data centers worldwide rely on water-based cooling as their primary method.

What Is Immersion Cooling?

Immersion cooling is a way of cooling IT components, including whole servers, by submerging them in a dielectric (electrically non-conducting) fluid.



The fluid has more than **1000 times** the thermal capacity of air (by volume), which makes immersion cooling a highly efficient alternative to air cooling. The liquid circulates by natural convection or is pumped to remove heat from the components.

In single-phase immersion cooling, the coolant stays as a liquid without phase changing, whereas the coolant boils to form a gas and then condenses back to a liquid in two-phase immersion cooling

The PROBLEM?

Even The Immersion Coolants Are Not Sustainable.

The PFAS Paradox

PFAS(Polyfluoro- alkyl substances), a class of approximately 12,000 compounds, are considered deadly due to their extreme persistence in the environment persisting in soil and water for **tens of thousands of years.**

For the same reason, PFAS face imminent regulatory **bans** from multiple countries.

In the context of **immersion cooling**, PFAS have been the **industry standard**, especially for double phase cooling.



The Miteni-Laxmi Organics case is arguably the most consequential PFAS story

-It involves the deliberate relocation of a PFAS manufacturing operation from a tightly regulated European jurisdiction to India, where no equivalent safeguards existed

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Miteni was an Italian chemical company that manufactured **PFAS** (“forever chemicals”).

Over many years, PFAS leaked from its plant in Trissino, Italy, contaminating groundwater and affecting **hundreds of thousands of people**.

In 2019, Laxmi Organic Industries bought Miteni’s machinery, patents, processes from **Italy to Lote Parshuram**, an industrial town in Maharashtra

It highlights global concerns about the continued production of “forever chemicals” despite growing evidence of their environmental and health impacts.

3M exiting PFAS has created the **largest** supply disruption in the history of liquid cooling, leaving the industry without its primary fluid source.

This creates **pressure** on other industries producing non-PFAS fluid, like synthetic hydrocarbon and mineral oils.

The BIGGER PROBLEM?

In Fact There's No Immersion Coolant Industry.
It's Oil and Chemicals Companies Selling By-products.

The Synthetic Supply Chain Story

Synthetic hydrocarbons are a type of single phase immersion coolant which occupy **42% of market share**

But the supply chain of these coolants is broken as it relies on borrowed capacity from two fragile industrial routes, PAO(Polyalphaolefin) and GTL(Gas to Liquid), that were **never designed** for the immersion coolant industry.

The three primary problems are:



Extreme Concentration: Just three companies control 91% of global PAO capacity, and only two companies worldwide license the technology needed for large-scale GTL production.

High Geographic Risk: Over 95% of US feedstock for PAO is concentrated in a single, hurricane-prone Gulf Coast corridor, while major GTL plants are located in geopolitical conflict zones.

Frequent Disruptions: This supply chain has suffered four major weather-related shutdowns in the US in under a decade, and the world's largest GTL plant in Qatar was recently damaged by military strikes.

THIS IS WHY THE WORLD NEEDS SYNTHETIC BIOLOGY BASED IMMERSION COOLANT

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We would be happy to take your QUESTIONS

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